

MAHESH THAMMAPPA

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SUMMARY

MCA student focused on Java backend and AI application development, with hands-on experience building Spring Boot microservices, REST APIs, and AI-powered applications. Skilled in integrating LLMs, vector databases, PostgreSQL, Spring Security, Docker, and CI/CD.

TECHNICAL SKILLS

Programming Language: Java

Core Java: OOP, Collections, Exception Handling, Multithreading, Generics, Java 8 Features

Frontend: HTML5, CSS3, Tailwind CSS, JavaScript, React.js

Backend: Spring Boot, Spring MVC, REST APIs, Spring Data JPA, Hibernate, Spring Security, JWT Authentication

Databases: SQL, PostgreSQL

Tools: Git, GitHub, Github Actions, Docker, Maven

AI/GenAI: LLM Integration, Embeddings, Vector Search, Google Gemini, pgvector, Pinecone

PROJECTS

PREPMATE: Intelligent Technical Interview Coach Using Vector Embeddings

Live | GitHub

- Developed an AI-powered interview platform that generates topic-based and resume-based technical interviews, evaluates candidate responses, and provides personalized feedback and model answers.
- Implemented semantic question deduplication using embeddings and Pinecone to prevent repetitive LLM-generated questions, along with a 70% mastery-based system that identifies weak and mastered concepts.
- Built a spaced-repetition practice workflow that stores low-scoring questions and generates new variations of weak concepts for continuous practice, using React, Java Spring Boot, PostgreSQL, Google Gemini, JWT/Spring Security, Docker, and GitHub Actions.

MEMORA: Personal Memory & Retrieval Platform

Live | GitHub

- Developed an AI-powered personal memory platform that allows users to record daily experiences, search past memories using natural language, and receive grounded responses with direct citations to the original journal entries.
- Implemented RAG-based memory retrieval using Gemini embeddings and PostgreSQL pgvector with cosine similarity search, along with AI-generated weekly reflections and a tamper-resistant 2-day journal locking mechanism to preserve chronological history.
- Built the application using React, Java Spring Boot, REST APIs, PostgreSQL, and Spring Security with JWT authentication; containerized the application using Docker, automated CI/CD with GitHub Actions, and deployed it on Render.

CERTIFICATIONS

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| • REST API (Intermediate) — HackerRank | Aug 2026 |
| • SQL (Intermediate) — HackerRank | Aug 2026 |
| • Software Engineering and Agile Software Development — Infosys | Mar 2026 |

EDUCATION

University Post Graduate College, Hyderabad

2024 – 2026

Master of Computer Applications (MCA)

Nizam College, Hyderabad

2020 – 2023

Bachelor of Science (Mathematics, Statistics & Computer Science)

CGPA: 8.14